

Delineating the Sorrow of Bihar: A Six-Channel U-Net for Semantic Segmentation of the August–September 2024 Kosi River Flood from Multi-Temporal Sentinel-1 SAR Imagery

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Abstract

The Kosi river, known locally as the Sorrow of Bihar for the frequency and severity of its avulsions, inundates a densely populated alluvial plain in northern Bihar, India, on a near-annual basis. Optical monitoring of these events is routinely defeated by monsoon cloud cover at the precise moment flood extent needs to be known, motivating the use of cloud-penetrating synthetic aperture radar (SAR). This paper reports on the design, training, and evaluation of a fully convolutional encoder-decoder network for pixel-wise flood segmentation over a $1.5^\circ \times 1.3^\circ$ reach of the Kosi basin, built from bi-temporal Sentinel-1 C-band SAR acquired before (1–20 August 2024) and after (1–20 September 2024) a significant flood pulse. A six-channel input stack – pre- and post-event VV and VH backscatter, a VV log-ratio change-detection layer, and SRTM elevation – was assembled in Google Earth Engine, exported at 10 m, tiled into 256×256 patches, and used to train a standard U-Net of 31.0 million parameters under a combined binary cross-entropy and Dice objective. After 87 epochs of training with early stopping, the network reached a test-set Dice coefficient of 0.9593 and an Intersection-over-Union of 0.9218, against 0.8129 Dice for a label-informed single-band change-detection threshold evaluated on the same test patches. We report the full training and evaluation protocol, including a phase-zero data audit, a comparison of median-filter and morphological-opening strategies for suppressing speckle-induced label noise, and a quantitative baseline comparison. We further discuss a methodological circularity inherent to threshold-derived reference labels – since one of the six input channels is the very quantity used to construct the ground truth – and its implications for interpreting the reported accuracy. A lightweight inference service exposing the trained model for single-patch prediction is also described.

Keywords: synthetic aperture radar, flood mapping, semantic segmentation, U-Net, Sentinel-1, Kosi river, Bihar, change detection, deep learning, remote sensing

1 Introduction

Few rivers in South Asia carry a reputation as fearsome as the Kosi. Rising in the Tibetan plateau and draining roughly $69,300 \text{ km}^2$ of the central Himalaya before it reaches the plains of northern Bihar, the Kosi has shifted its course westward by more than 120 km over the past two and a half centuries, a migration driven by the extraordinary sediment load it carries down from young, tectonically active mountains [14]. Successive attempts to fix the river in place with embankments – built after the 1954 floods and completed by 1963 – have repeatedly failed: major breaches occurred in 1963, 1968, 1971, 1980, 1984, 1987, 1991, and, most catastrophically, on 18 August 2008, when the eastern afflux bund gave way near Kusaha in Nepal and sent the river down a channel it had not occupied in generations, displacing over three million people in five Bihar districts [16, 15]. It is this history that earned the Kosi its Bihari epithet: the Sorrow of Bihar.

The flood-prone area of Bihar grew from roughly 2.5 million hectares in 1954 to 6.8 million hectares by the mid-1990s [16], and the Kosi basin alone accounts for a disproportionate share of that expansion. Every monsoon season since has brought a renewed round of inundation to the same low-lying, densely

farmed districts, and every season the same operational question recurs: where, precisely, is the water, and how much land does it cover at a given moment. Answering that question quickly enough to be useful for evacuation planning, relief logistics, or post-event damage assessment has historically depended on ground survey and, more recently, on optical satellite imagery. Both are limited in the same way – ground survey is slow and hazardous during an active flood, and optical sensors are blind exactly when floods are most active, because monsoon flooding and monsoon cloud cover are the same meteorological event viewed from two different altitudes.

Synthetic aperture radar sidesteps this limitation by illuminating the ground with its own microwave energy rather than relying on reflected sunlight, so that acquisition is unaffected by cloud, haze, or the time of day. The Copernicus Sentinel-1 mission, operated by the European Space Agency, has provided freely available C-band SAR imagery with sub-weekly revisit over most of the globe since 2014 [1], and its Ground Range Detected (GRD) product in Interferometric Wide swath mode has become a de facto standard input for operational flood mapping services. Open water is a comparatively strong specular reflector at C-band incidence angles typical of Sentinel-1’s IW mode: smooth water surfaces redirect the bulk of the incident energy away from the sensor, so flooded ground appears as a region of markedly depressed backscatter relative to its dry-season state. This physical contrast is the basis of essentially every SAR flood-mapping method published over the last two decades, from single-image amplitude thresholding through bi-temporal change detection to the deep convolutional networks that are now the dominant approach in the research literature [13].

The work reported here applies that general strategy to a single, well-documented event: the flood pulse that moved down the Kosi basin between the closing weeks of August and the middle of September 2024. Rather than relying on a hand-labelled, expert-curated reference dataset – the norm in the benchmark literature, and a genuine bottleneck for site-specific studies – this study constructs its training labels directly from the bi-temporal SAR signal itself, using a change-detection threshold refined against the pixel statistics of the study area and cross-checked with a permanent-water mask so that perennial river channels and oxbow lakes are not misclassified as flood. A standard U-Net encoder-decoder, taking as input a six-channel stack of pre- and post-event backscatter, the derived change-detection layer, and terrain elevation, is then trained to reproduce this reference at the pixel level. The resulting model is evaluated against a held-out test partition and against a non-learned threshold baseline evaluated under identical conditions, and the paper devotes particular attention to the methodological consequences of deriving both the reference labels and one of the model’s input channels from the same underlying quantity – a circularity that is common in the self-supervised SAR flood-mapping literature but is not always made explicit.

The remainder of the paper is organised as follows. Section 2 surveys prior work on SAR-based flood detection, from classical thresholding through the deep learning architectures now dominant in the field, and situates the present study relative to that literature. Section 3.1 describes the Kosi study area, the Sentinel-1 acquisitions used, and the full processing pipeline, from Earth Engine compositing through patch extraction, label refinement, network architecture, and training protocol. Section 4 reports quantitative and qualitative results, compares the trained network against a non-learned threshold baseline, and discusses the findings, their limitations, and a methodological circularity in the reference-label construction. Section 5 concludes and outlines directions for future work.

2 Literature Survey

2.1 SAR backscatter physics and threshold-based flood detection

The radiometric basis for SAR flood mapping is well established: calm open water behaves close to a specular reflector at the incidence angles used by spaceborne SAR systems, scattering most of the incident microwave energy away from the sensor and producing backscatter coefficients that are markedly and consistently lower than those of vegetated, bare, or built-up land [13]. A single post-event SAR

image can therefore, in principle, be segmented into water and non-water classes by applying a global or locally adaptive intensity threshold, and this remains the conceptual starting point for the great majority of subsequent methods. Histogram-based automatic thresholding, most commonly Otsu’s method [4], which selects the threshold that maximises inter-class variance under an implicit bimodality assumption, has been applied extensively to single-date SAR amplitude images for water delineation. Its principal weakness is well documented: threshold selection degrades sharply when the water class occupies a small, non-dominant fraction of the scene, when double-bounce scattering from flooded vegetation or inundated urban structures raises backscatter back into the land-like range, or when speckle noise broadens both class distributions until they overlap. Liang and Liu [5] addressed the small-water-fraction failure mode directly with a locally adaptive thresholding scheme that partitions the scene into subregions before applying Otsu’s criterion, improving robustness where flood extent is a minority class, a scenario that, as will be seen in Section 4, closely matches the class balance encountered in the present study area.

Bi-temporal change detection improves on single-date thresholding by comparing a post-event acquisition against a pre-event reference over the same geometry, forming either a ratio or a log-ratio image that suppresses much of the terrain- and incidence-angle-dependent backscatter variation that would otherwise confound a single-date threshold. Flooded pixels manifest as a pronounced negative excursion in the log-ratio layer, and a threshold applied to that derived layer – rather than to raw amplitude – is markedly more robust to the underlying land cover. This change-detection formulation, applied here to the VV log-ratio between pre- and post-flood Sentinel-1 composites, is the same conceptual device used to construct the reference labels described in Section 3.3, and it is also the formulation adopted for the non-learned baseline against which the trained network is compared in Section 4.

2.2 Deep learning for SAR flood segmentation

The limitations of thresholding – sensitivity to speckle, confusion with double-bounce and layover terrain, and the need for scene-specific tuning – motivated a shift toward supervised, spatially aware classifiers over the last decade. The U-Net architecture, originally proposed by Ronneberger et al. [6] for biomedical image segmentation, has become the dominant backbone in this literature. Its symmetric encoder-decoder structure, in which a contracting path extracts increasingly abstract, low-resolution features while a matched expanding path recovers full-resolution output, and skip connections carry high-resolution spatial detail directly from encoder to decoder, is particularly well suited to flood mapping, where the model must simultaneously reason about broad inundation context and preserve the fine, often linear boundaries of individual channels and floodplain margins.

The largest catalyst for supervised SAR flood segmentation research has been the release of Sen1Floods11 by Bonafilia et al. [8], a georeferenced benchmark of 4,831 Sentinel-1 and Sentinel-2 chips spanning eleven global flood events, of which 446 carry hand-corrected, expert-verified labels. Bonafilia et al. demonstrated that fully convolutional networks trained on this benchmark outperform threshold-based baselines by a wide margin, a finding that has been repeatedly reproduced in the years since. Konapala et al. [9] extended this line of work by combining Sentinel-1 and Sentinel-2 bands as U-Net input and, notably for the present study, showed that appending a digital elevation model as an auxiliary channel measurably improves segmentation accuracy when only SAR imagery is otherwise available – a result that directly motivated the inclusion of an SRTM elevation band in the six-channel input stack used here. Paul and Ganju [10] explored semi-supervised training on Sentinel-1 alone, using ensembles of U-Net variants to generate pseudo-labels on unlabelled imagery, addressing the same scarcity of hand-annotated SAR flood labels that motivates the threshold-derived label strategy adopted in this paper.

More recent architectural work has pursued refinements to the base U-Net rather than wholesale replacement. Jamali et al. [11] introduced a Residual Wave Vision U-Net that augments a ResU-Net backbone with vision-MLP token mixing over dual-polarisation Sentinel-1 features, reporting precision above 95% on their evaluation set. Ghosh et al. [12] evaluated a nested U-Net (U-Net++) against a NASA

benchmark flood dataset built from Sentinel-1 imagery, and comparative studies elsewhere in the literature have begun to weigh U-Net variants against vision-transformer segmentation heads, reporting that transformer self-attention improves the capture of long-range spatial context at the cost of substantially higher data and compute requirements relative to a convolutional encoder-decoder. Amitrano et al. [13] provide a comprehensive recent review of SAR flood detection techniques and datasets, and note that despite the proliferation of architectures, a large share of published studies continue to report accuracy on curated, often globally aggregated benchmarks rather than on single, site-specific events of the kind addressed here.

2.3 Ancillary data and label generation

Because flooded and permanently inundated pixels are radiometrically indistinguishable at a single SAR wavelength, essentially all flood-mapping pipelines – learned or otherwise – require an auxiliary permanent-water mask to isolate the transient, event-specific inundation signal. The JRC Global Surface Water dataset compiled by Pekel et al. [2] from more than three million Landsat scenes remains the most widely used resource for this purpose, and its monthly water-occurrence layer is used in the present study, in Google Earth Engine, to exclude perennial river channels and lakes from the flood reference label. Terrain elevation, most commonly sourced from the Shuttle Radar Topography Mission [3], serves a complementary role: it both supplies the auxiliary input channel found beneficial by Konapala et al. [9] and, in the present pipeline, is used to mask out steep, hilly terrain where SAR-based flood detection is known to be unreliable owing to layover and shadow.

2.4 Positioning of the present study

The literature summarised above is overwhelmingly evaluated on multi-event, multi-region benchmarks such as Sen1Floods11, which offer strong generalisation evidence but necessarily average over considerable heterogeneity in land cover, hydrology, and SAR acquisition geometry. Comparatively few studies report a deep segmentation model trained and evaluated end-to-end on a single, densely characterised event over the Kosi basin specifically, despite the basin’s outsized humanitarian relevance in South Asia. This paper contributes such a study: a complete, auditable pipeline from raw Sentinel-1 GRD acquisition through label construction, patch-level dataset curation, network training, and quantitative baseline comparison, applied to the August–September 2024 Kosi flood pulse. It also departs from much of the surveyed literature in making explicit – rather than leaving implicit – the fact that its reference labels are threshold-derived rather than independently surveyed, and in quantifying, through the baseline comparison of Section 4, how much of the network’s apparent skill can be attributed to that shared derivation.

3 Methodology

3.1 Study area

The study area is a rectangular region of northern Bihar bounded by 86.0°E to 87.5°E and 25.5°N to 26.8°N, chosen to encompass the active Kosi channel belt and its surrounding floodplain within the districts historically most exposed to Kosi flooding. The region is characterised by low relief, dense agricultural land use, a high density of seasonal and perennial water bodies, and a braided, highly mobile river channel that itself constitutes a substantial fraction of the permanent-water class within the scene.

3.2 Sentinel-1 acquisition and pre-processing

Sentinel-1 Ground Range Detected imagery was retrieved via Google Earth Engine [17] from the COPERNICUS/S1_GRD collection [18], filtered to Interferometric Wide (IW) swath mode, descending orbit pass, and dual VV/VH polarisation. Two 20-day compositing windows were defined: a pre-flood window from 1–20 August 2024 and a post-flood window from 1–20 September 2024, each reduced to a per-pixel temporal mean to suppress residual speckle prior to change detection. Both composites were clipped to the study

rectangle.

A change-detection layer was then formed as the pixel-wise difference, in decibels, between the post- and pre-flood VV composites,

$$L = \sigma_{VV,post}^0 - \sigma_{VV,pre}^0, \quad (1)$$

so that flooded pixels, which lose backscatter after inundation, present as negative values of L (referred to throughout as the log-ratio band) while stable land cover clusters near zero. Descriptive statistics of L over the study rectangle gave a mean of -0.08 dB and a standard deviation of approximately 2.96 dB; a flood threshold of -3 dB, close to one standard deviation below the mean, was adopted for reference-label construction after an initial unsupervised Otsu threshold on the full-scene histogram was found to be uninformative, for reasons discussed in Section 3.11.

Permanent water was identified from the JRC Global Surface Water seasonality band [2, 19], with pixels present for at least ten months of the year classified as permanently inundated and excluded from the flood label, so that the reference layer captures event-driven inundation rather than the river’s ordinary wetted channel. Terrain elevation was sourced from the 30 m SRTM digital elevation model [3, 20]; a flat-terrain mask retaining only pixels below 200 m elevation was applied throughout, since the study region is entirely lowland floodplain and this threshold conservatively removes any hillslope terrain at the rectangle’s margins where radar layover and shadow would otherwise compromise the change-detection signal.

The final export stack comprises six bands at 10 m spatial resolution and geographic (EPSG:4326) projection: post-flood VV, post-flood VH, pre-flood VV, pre-flood VH, the VV log-ratio of Equation 1, and SRTM elevation. Table 1 summarises the input stack, and the corresponding reference flood label – the thresholded, permanent-water-masked log-ratio layer – was exported as a co-registered single-band raster at identical resolution and extent.

Table 1: Six-channel SAR input stack exported from Google Earth Engine.

Band	Symbol	Description
1	VV_post	Post-flood VV backscatter, 1–20 Sep 2024 mean composite
2	VH_post	Post-flood VH backscatter, 1–20 Sep 2024 mean composite
3	VV_pre	Pre-flood VV backscatter, 1–20 Aug 2024 mean composite
4	VH_pre	Pre-flood VH backscatter, 1–20 Aug 2024 mean composite
5	log_ratio	VV change-detection layer, Eq. 1
6	elevation	SRTM digital elevation model, metres

Owing to Earth Engine export tiling, the SAR stack was delivered as four adjoining GeoTIFF tiles rather than a single raster; these were subsequently mosaicked (Section 3.3) into a virtual raster spanning $16,699 \times 14,495$ pixels at full resolution, alongside a single-tile label raster of matching extent.

3.3 Data processing pipeline

The processing pipeline comprises five stages: a phase-zero data audit, virtual mosaicking and patch extraction, reference-label refinement, network training, and evaluation against a non-learned baseline. Each is described in turn.

3.4 Phase zero: data audit

Before any model development, the four SAR tiles and the label raster were audited for internal consistency and data quality. Each tile’s dimensions, coordinate reference system, band count, and bounding box were verified against the expected mosaic geometry, and windowed reads at each tile’s four corners

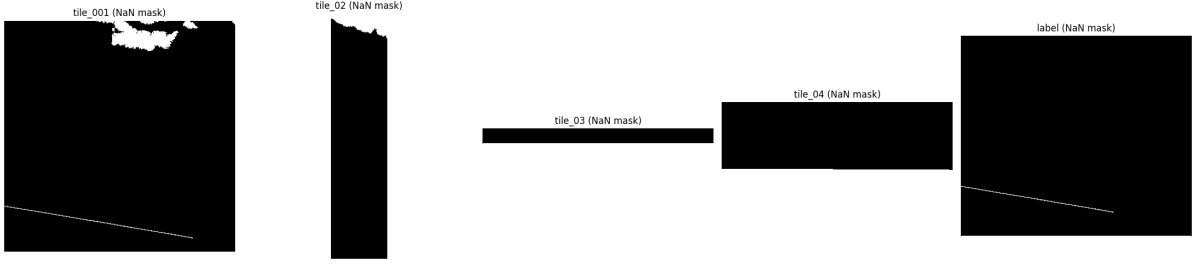


Figure 1: No-data footprint (white) across the four SAR tiles and the label raster, derived from a decimated read of each raster at reduced resolution. No-data regions are concentrated at tile margins rather than distributed through the interior, confirming that the four-tile mosaic can be virtually stitched without interior data loss.

were used to establish that no-data regions, where present, occurred as edge artefacts rather than interior gaps that would silently corrupt patch extraction.

A block-wise scan of the full-resolution rasters, reading 1024×1024 -pixel windows sequentially to keep peak memory use bounded regardless of file size, quantified the no-data fraction per tile: 1.94%, 5.96%, 1.71%, and 0.76% for the four SAR tiles respectively, and 0.16% for the label raster (Figure 1). The same block-wise strategy was used to compute the global class balance of the reference label prior to patching: of 241,664,048 valid label pixels within the study rectangle, 31,991,717 (13.24%) were flagged as flood, establishing at the outset that flood is a substantial minority class at the pixel level, a property that persists into the patch-level dataset described next and that motivates the choice of a combined cross-entropy and Dice loss in Section 3.8.

3.5 Virtual mosaicking and patch extraction

The four SAR tiles were combined into a single virtual raster (VRT) using GDAL’s `BuildVRT` utility with nearest-neighbour resampling, avoiding duplication of pixel data on disk while presenting a single, contiguous $16,699 \times 14,495$ -pixel raster to downstream code. A non-overlapping sliding window of 256×256 pixels (patch size equal to stride, so no patch shares pixels with its neighbours) was then swept across the mosaic in a regular grid, yielding 57 row and 66 column positions and 3,762 candidate patch locations.

At each candidate location, the corresponding six-band SAR window and single-band label window were read directly from the VRT and label raster respectively. A patch was retained only if at least 70% of its pixels were valid (non-NaN) in the first SAR band, discarding locations dominated by the tile-edge no-data artefacts identified in the phase-zero audit; the remaining NaN values within retained patches were filled with zero. Retained SAR patches were cast to `float16` and retained label patches to `uint8` before being written to disk as individual `.npy` arrays, keeping the on-disk footprint of the full patch archive within the storage budget of the training environment. This procedure yielded 3,652 patches surviving the validity filter out of 3,762 candidate grid cells (Figure 2).

3.6 Reference label refinement

Inspection of individual retained patches (Figure 2) showed that the raw thresholded label, while radiometrically correct in aggregate, carried substantial pixel-level speckle: isolated flood-flagged pixels scattered through otherwise dry terrain, and isolated dry-flagged holes within otherwise flooded regions, consistent with the well-documented multiplicative speckle noise inherent to SAR amplitude imagery rather than genuine sub-pixel inundation structure.

Two denoising strategies were compared on representative low-, medium-, and high-density patches: a 5×5 median filter, and, separately, a 3×3 median filter and a 2×2 binary morphological opening evaluated side by side (Figures 3 and 4). The 5×5 median filter was found to erode genuine flood structure disproportionately at the smaller patch scale, thinning narrow inundated channels to the point of

disconnection. The 2×2 morphological opening – an erosion immediately followed by a dilation with the same 2×2 structuring element – removed isolated single-pixel speckle while leaving contiguous flood regions comparatively intact, and was adopted as the final label-cleaning operation applied uniformly across all 3,652 retained patches. The resulting opened label is referred to throughout the remainder of this paper as the reference (ground-truth) label.

Table 2 summarises the effect of each candidate operator on the mean per-patch flood fraction across the archive. The morphological opening reduced the mean flood fraction only marginally relative to the unfiltered label, consistent with the removal of isolated speckle rather than the erosion of contiguous structure, whereas patches reduced to zero flood pixels after filtering were more frequent under the 5×5 median filter than under the 2×2 opening.

Table 2: Effect of candidate label-cleaning operators on per-patch flood fraction across all 3,652 retained patches.

Label variant	Mean flood fraction	Patches driven to zero flood pixels
Unfiltered (raw threshold)	higher	fewer
5×5 median filter	lower	more
2×2 morphological opening (adopted)	comparable to unfiltered	fewer

3.7 Dataset partitioning

The 3,652-patch archive was split into training, validation, and test partitions using a two-stage stratified split keyed on a discretised flood-density bin so that each partition preserves a comparable distribution of flood-fraction extremes rather than being dominated by low-density patches by chance. A held-out test partition of 15% was separated first, followed by a further 15%-of-remainder split into training and validation subsets, both performed with a fixed random seed for reproducibility. This produced 2,638 training patches, 466 validation patches, and 548 test patches. The three partitions carry near-identical mean per-patch flood fraction (train: 0.0921, std 0.0497; validation: 0.0906, std 0.0484; test: 0.0921, std 0.0488), confirming that the stratified split achieved its intended balance. The overall archive mean flood fraction, after the morphological-opening label cleanup, is 0.0919.

3.8 Network architecture and loss function

The segmentation network is a standard U-Net [6], adapted to a six-channel input to match the SAR stack described in Table 1 and a single-channel output representing per-pixel flood logits. The encoder comprises four downsampling stages at feature widths of 64, 128, 256, and 512 channels, each stage a pair of 3×3 convolutions with batch normalisation and ReLU activation (a DoubleConv block) followed by 2×2 max pooling. A bottleneck DoubleConv block doubles the final encoder width to 1,024 channels. The decoder mirrors the encoder with four upsampling stages, each a 2×2 transposed convolution followed by channel-wise concatenation with the corresponding encoder skip connection and a DoubleConv block, before a final 1×1 convolution reduces the output to a single flood-probability logit per pixel. The complete network totals 31,045,249 trainable parameters.

The training objective combines binary cross-entropy with a differentiable Dice loss in equal proportion,

$$\mathcal{L} = 0.5 \mathcal{L}_{\text{BCE}}(\hat{y}, y) + 0.5 \left(1 - \frac{2 \sum_i \hat{y}_i y_i + \epsilon}{\sum_i \hat{y}_i + \sum_i y_i + \epsilon} \right), \quad (2)$$

where $\hat{y}$ is the sigmoid-activated prediction, y the reference label, and $\epsilon = 10^{-6}$ a smoothing constant preventing division by zero on flood-free patches. This combination, following Milletari et al. [7], was adopted to counteract the class imbalance quantified in Section 3.1 without resorting to explicit class reweighting: cross-entropy provides a well-behaved per-pixel gradient early in training, while the Dice term directly optimises the overlap metric used for evaluation and is comparatively insensitive to the dominance of the non-flood class.

3.9 Training protocol

The network was trained with the Adam optimiser at an initial learning rate of 1×10^{-4} , reduced by a factor of 0.5 whenever validation loss failed to improve for three consecutive epochs (ReduceLROnPlateau). Training used a batch size of 16 with data loading restricted to the main process, since parallel data-loader workers were found to conflict with the read-only mounted storage of the training environment. Training was capped at 200 epochs with early stopping on validation Dice at a patience of 15 epochs, and the checkpoint with the highest validation Dice was retained for final evaluation. Training was executed on a pair of NVIDIA Tesla T4 GPUs under DataParallel, with a mean epoch time of approximately 107 seconds.

In practice, training terminated after 87 epochs, having last improved validation Dice at epoch 72 (val Dice 0.9601) and shown no further improvement for the subsequent 15 epochs. Figures 5, 6, and 7 show the resulting training and validation loss, Dice, and IoU trajectories respectively. The learning-rate schedule stepped down five times over the course of training (visible as the successive plateaus and drops in the validation curves), and validation Dice exhibits the characteristic sawtooth of a scheduler responding to transient plateaus rather than a smooth monotonic climb, with brief regressions around epochs 15, 25, 37, 52, and 63 that recovered within a small number of subsequent epochs.

3.10 Evaluation metrics

Test-set performance is reported using the Dice coefficient, Intersection-over-Union (Jaccard index), precision, recall, F1 score, and specificity, all computed at the pixel level over the full test partition after thresholding the sigmoid-activated network output at 0.5. Precision, recall, and F1 were computed using the binary-average convention over the flood class; specificity was computed as $TN/(TN + FP)$ from the pixel-level confusion matrix.

3.11 Non-learned baseline

To contextualise the trained network’s performance, three variants of a single-band threshold classifier operating on the log-ratio channel alone were evaluated under identical test-set conditions. The first applied Otsu’s method independently to each test patch’s local log-ratio histogram; the second applied a single global Otsu threshold estimated once from the pooled training-set log-ratio distribution and then held fixed across the test set; the third applied a label-informed threshold, obtained by maximising Youden’s J statistic ($TPR - FPR$) over an ROC analysis of training-set log-ratio values against the training reference labels. The three variants isolate, respectively, the effect of per-patch adaptivity, of a single scene-wide statistic, and of a threshold chosen with full knowledge of the reference labels – the last representing the strongest baseline a purely threshold-based method could plausibly achieve on this dataset, and the one reported as the primary comparison point in Section 4.

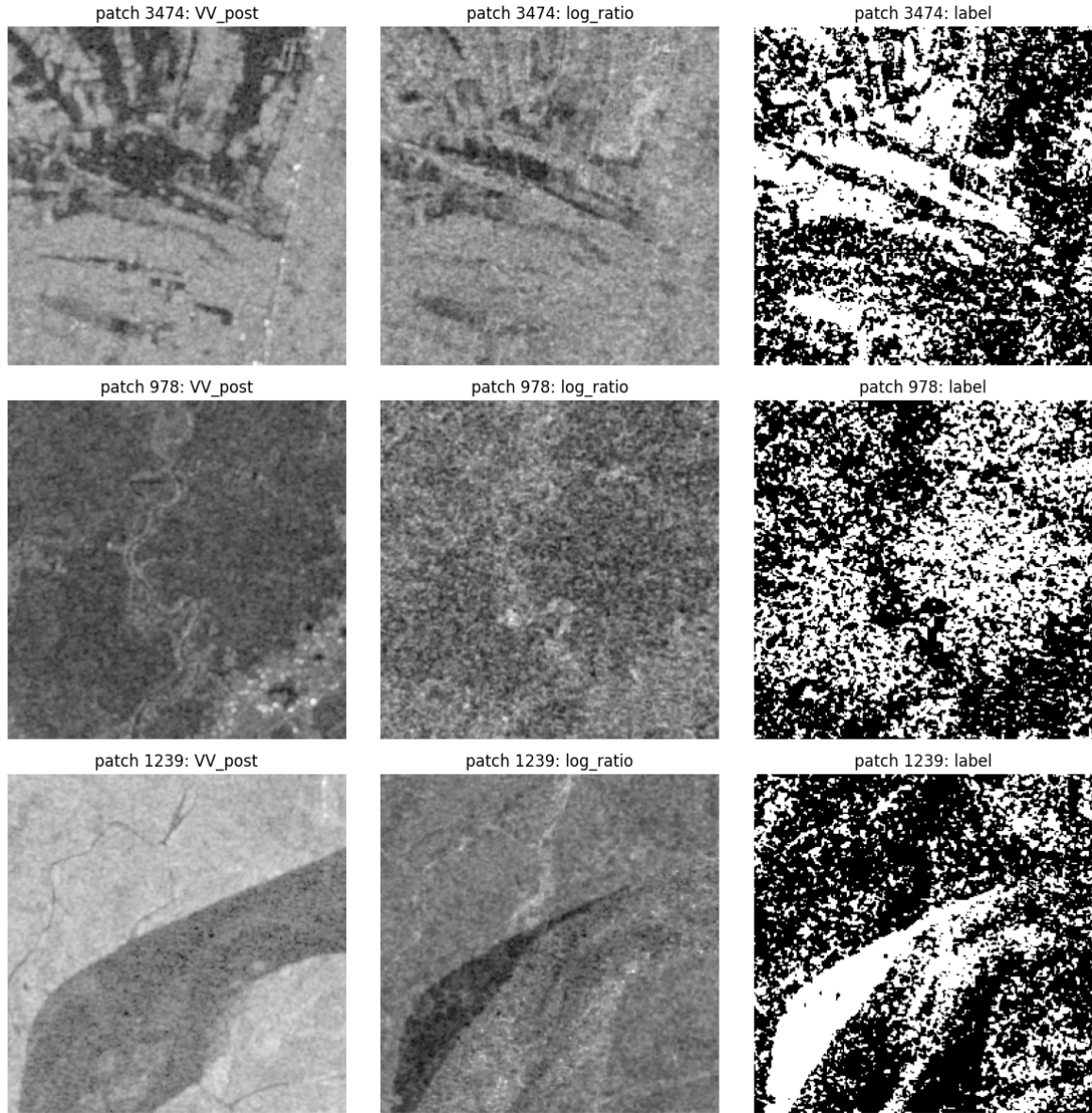


Figure 2: Three sample 256×256 training patches, selected from among the highest flood-density patches by pixel count, showing (left to right) post-flood VV backscatter, the VV log-ratio change-detection layer, and the corresponding reference flood label. Flooded regions manifest as darker, lower-backscatter zones in VV_post and as diffuse negative excursions in the log-ratio layer that the reference label captures as speckled point clusters rather than smooth, contiguous regions.

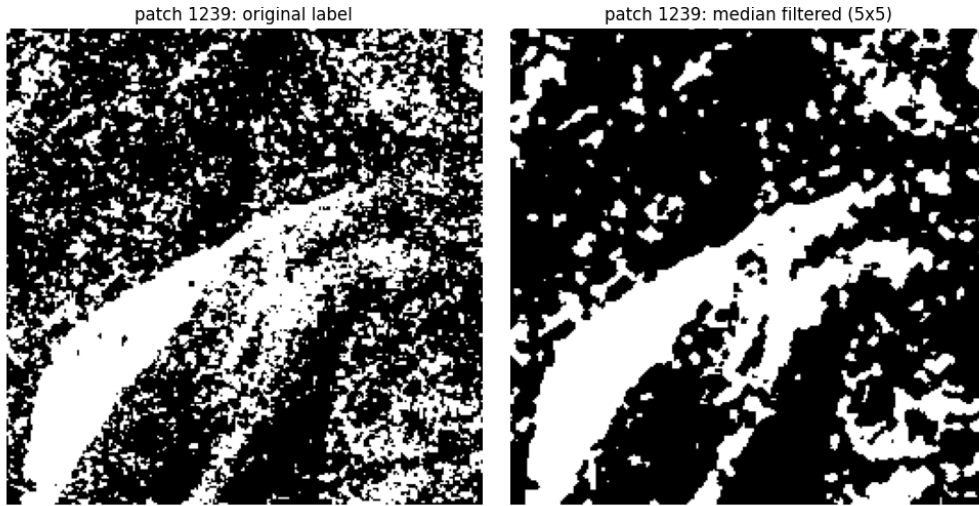


Figure 3: Effect of a 5×5 median filter on a representative reference-label patch. The filtered label (right) suppresses isolated speckle but visibly thins narrow flood structures relative to the unfiltered original (left), motivating the comparison of alternative filter sizes and morphological operators in Figure 4.

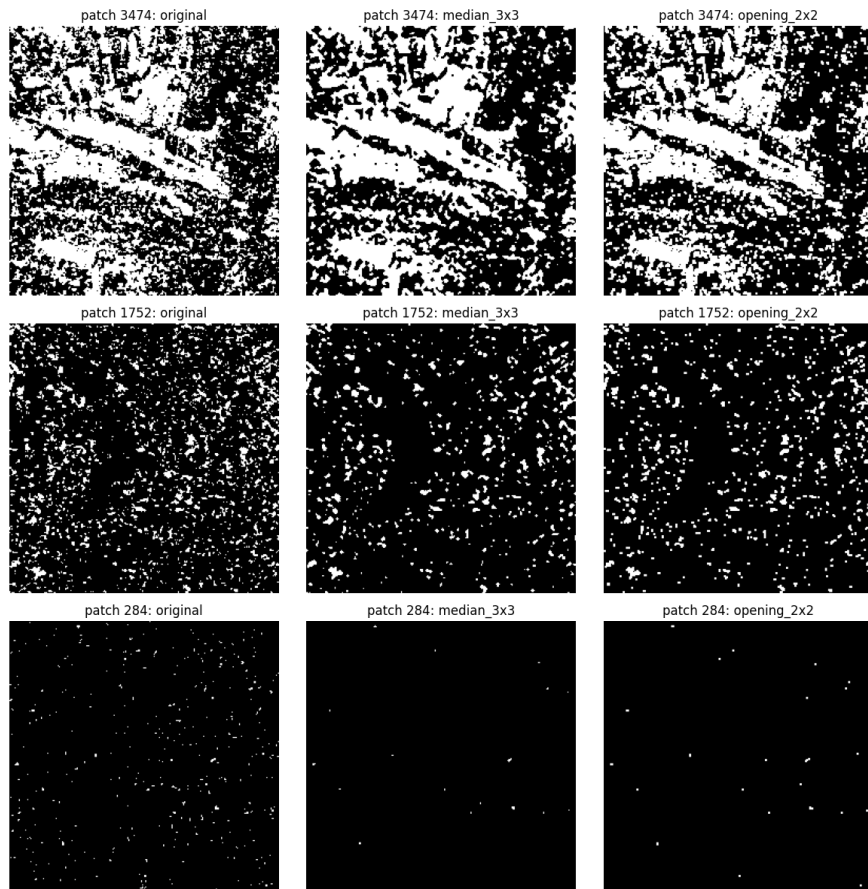


Figure 4: Comparison of the original reference label against a 3×3 median filter and a 2×2 binary morphological opening, evaluated on low-, medium-, and high-density flood patches. The morphological opening was adopted as the final label-cleaning operator applied across the full patch archive.

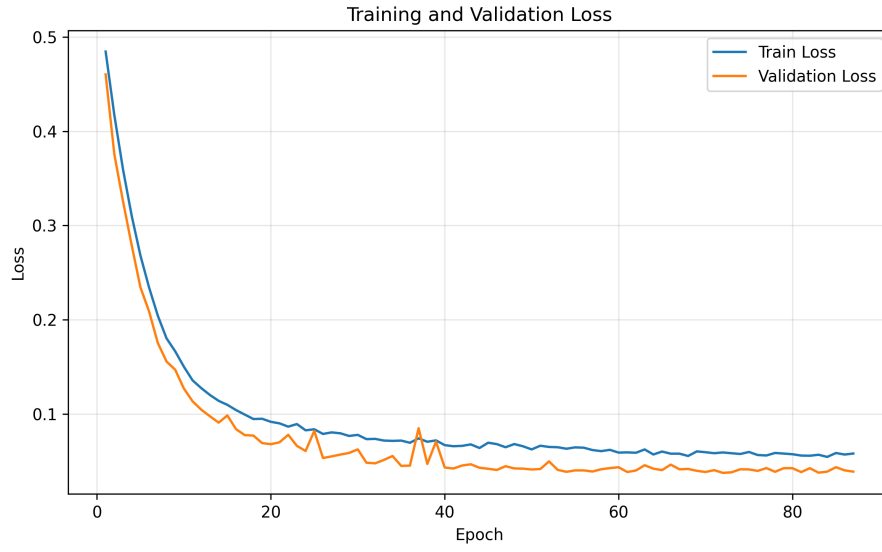


Figure 5: Training and validation combined BCE-Dice loss over 87 epochs. The best checkpoint, used for all test-set evaluation reported in Section 4, was saved at epoch 72.

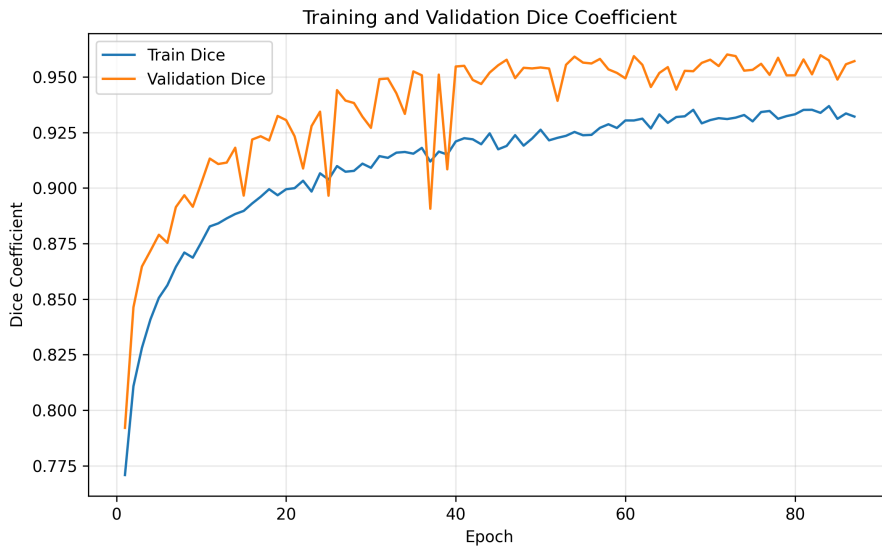


Figure 6: Training and validation Dice coefficient over 87 epochs.

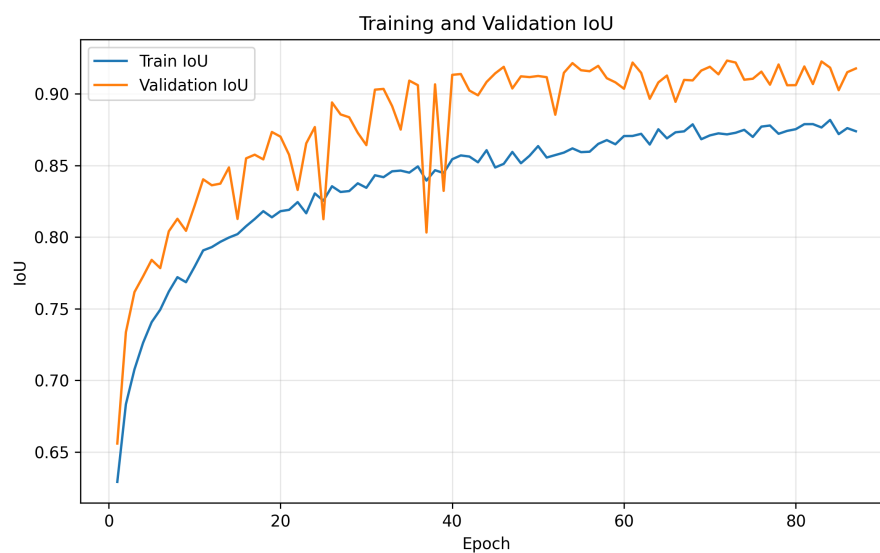


Figure 7: Training and validation Intersection-over-Union over 87 epochs.

4 Results and Discussion

4.1 Overall test-set performance

Table 3 reports the trained network’s performance on the 548-patch held-out test partition, evaluated from the checkpoint saved at epoch 72. The network attained a Dice coefficient of 0.9593 and an IoU of 0.9218, with precision and recall both above 0.959 and a specificity of 0.9959, indicating that the great majority of the non-flood class is correctly rejected alongside strong recovery of the flood class itself.

Table 3: Test-set performance of the trained U-Net, best checkpoint (epoch 72), $n = 548$ patches.

Metric	Value
Test loss (BCE + Dice)	0.0387
Dice coefficient	0.9593
Intersection-over-Union	0.9218
Precision	0.9591
Recall	0.9600
F1 score	0.9596
Specificity	0.9959

The pixel-level confusion matrix underlying Table 3 is $TN = 32,502,727$, $FP = 134,041$, $FN = 130,979$, $TP = 3,145,981$, and its row-normalised form is shown in Figure 8. False positives and false negatives are close to symmetric in absolute count, with neither class showing a pronounced systematic bias, though the flood class – being the minority class at roughly 8.8% of test-set pixels – carries a substantially higher relative error rate than the dominant non-flood class, an asymmetry consistent with the class imbalance documented in Section 3.1.

Figure 9 presents the same set of metrics, together with overall pixel accuracy (0.9925), as a summary bar chart, and Figure 10 shows the distribution of the log-ratio band separately for flood- and non-flood-labelled pixels drawn from a subsample of the training partition. The two distributions are clearly separated in central tendency but carry substantial overlapping tails, visually confirming why a single global threshold on this band alone (the baseline classifier of Section 3.11) cannot achieve the precision reached by the multi-band, spatially aware network.

4.2 Qualitative segmentation results

Figures 11, 12, and 13 show representative predictions on test patches drawn from the low, median, and highest flood-density strata of the test partition respectively, alongside the corresponding post-flood VV backscatter, log-ratio layer, and reference label. At low flood density the network correctly suppresses flood predictions across the great majority of the patch, with only isolated small false positives. At median density (Figure 12), a scattered, speckle-like reference-label pattern – itself an artefact of the change-detection labelling process rather than of genuinely disjoint flood patches – is closely reproduced by the network, including much of its speckled spatial structure, suggesting that the network has learned to approximate not only the underlying inundation signal but also the particular noise characteristics of the labelling procedure. At the highest flood density observed in the test partition (Figure 13), the network correctly identifies the great majority of the flood extent, with prediction boundaries closely tracking the reference label’s irregular, dendritic flood-front geometry.

4.3 Comparison against the non-learned baseline

Table 4 reports the three threshold-baseline variants described in Section 3.11. The per-patch Otsu variant performed poorly (Dice 0.2904), driven by a very low precision of 0.1699 against a perfect recall of 1.0000: with the flood class occupying only a small minority of most patches, per-patch Otsu thresholding tends to bisect the local histogram in a way that over-predicts flood area substantially, a known failure mode of unsupervised thresholding under severe class imbalance and one of the motivations cited

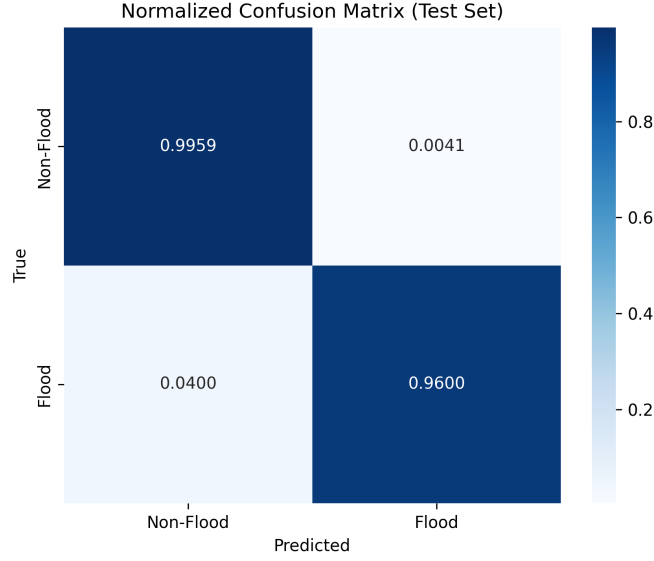


Figure 8: Row-normalised pixel-level confusion matrix on the test partition. Rows sum to 1 and represent the true class; columns represent the predicted class.

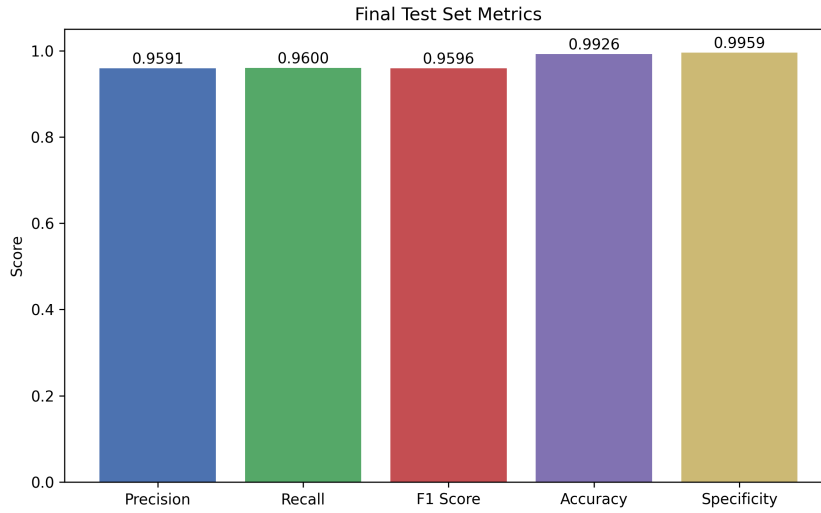


Figure 9: Summary of test-set precision, recall, F1 score, accuracy, and specificity for the trained network.

by Liang and Liu [5] for locally adaptive rather than purely local thresholding schemes. The global Otsu variant performed comparably poorly (Dice 0.2676), with an estimated threshold of $+0.24$ dB that, applied uniformly, again drives recall to 1.0000 at the cost of precision (0.1545), since a threshold optimised over the full log-ratio histogram – dominated by the non-flood majority class – does not correspond to a threshold that discriminates the minority flood class well.

The label-informed Youden’s J threshold, at -3.00 dB, coincides closely with the -3 dB threshold used to construct the reference labels themselves (Section 3.1), which is expected given that both are derived from closely related statistics of the same log-ratio distribution. This baseline achieved a substantially stronger Dice of 0.8129 and IoU of 0.6848, with near-perfect recall (0.9992) but moderate precision (0.6852), reflecting that a single fixed threshold on log-ratio alone, even when chosen optimally with full label knowledge, cannot achieve the spatial precision of a network that additionally exploits the VH channels, absolute pre- and post-event backscatter, elevation, and, crucially, spatial context across the 256×256 receptive field. Figure 14 visualises this comparison directly against the trained network across five shared metrics.

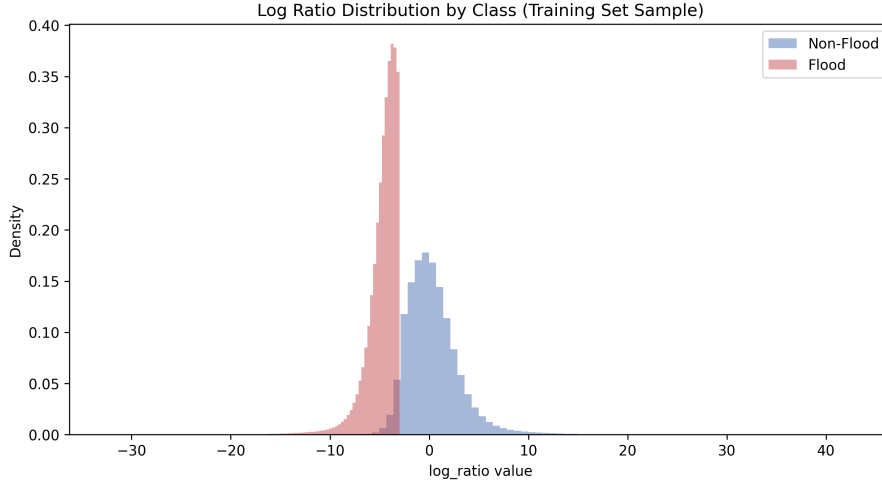


Figure 10: Distribution of the VV log-ratio change-detection band for flood- and non-flood-labelled pixels, sampled from the training partition. The classes are well separated in mean but overlap substantially in their tails, which bounds the achievable accuracy of any classifier that considers the log-ratio band in isolation.

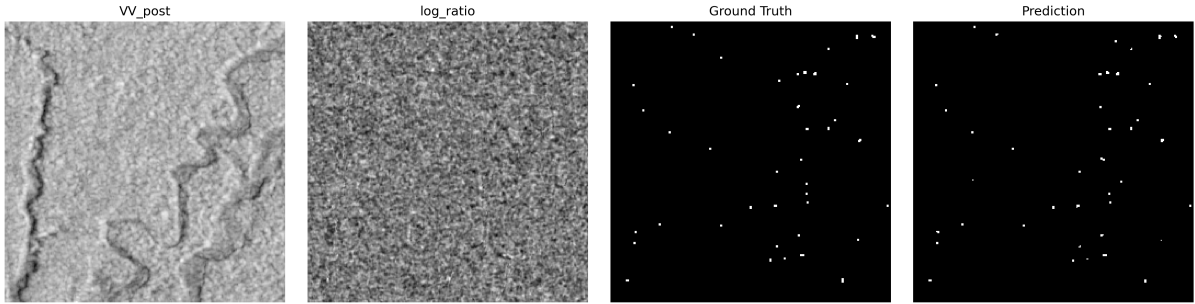


Figure 11: Qualitative comparison on the lowest flood-density test patch: post-flood VV backscatter, log-ratio, reference label, and network prediction (left to right).

4.4 Interpreting the accuracy gap over the threshold baseline

The roughly 15-point Dice improvement of the trained network over the strongest, label-informed threshold baseline (0.9593 versus 0.8129) is substantial, but attributing it entirely to the network’s representational capacity would overstate the case. Three factors plausibly available to the network but structurally unavailable to the single-band threshold are worth separating. First, spatial context: because the network’s receptive field spans the greater part of a 256×256 patch, it can exploit local continuity of flood boundaries and regularities of channel geometry that a per-pixel threshold cannot see. Second, additional radiometric information: the VH channels and the raw pre- and post-event VV amplitudes (rather than only their difference) carry information the log-ratio band alone discards, particularly in double-bounce or layover-affected pixels where the ratio signal is ambiguous but the absolute polarimetric response may not be. Third, and most subtly, the network was trained to reproduce a reference label that had itself been denoised by 2×2 morphological opening (Section 3.3); to the extent that the network internalises this denoising operation as part of its learned mapping, part of its apparent accuracy gain over a raw per-pixel threshold reflects the network reproducing a smoothing step that was applied to its own training target rather than reflecting an improvement in the underlying physical detection of flood water.

4.5 The label-derivation circularity

A limitation that merits explicit statement, since it bears directly on how the headline Dice and IoU figures should be read, is that the reference labels used both for training and for test-set evaluation were themselves constructed from a threshold applied to the log-ratio band – the same band that is supplied

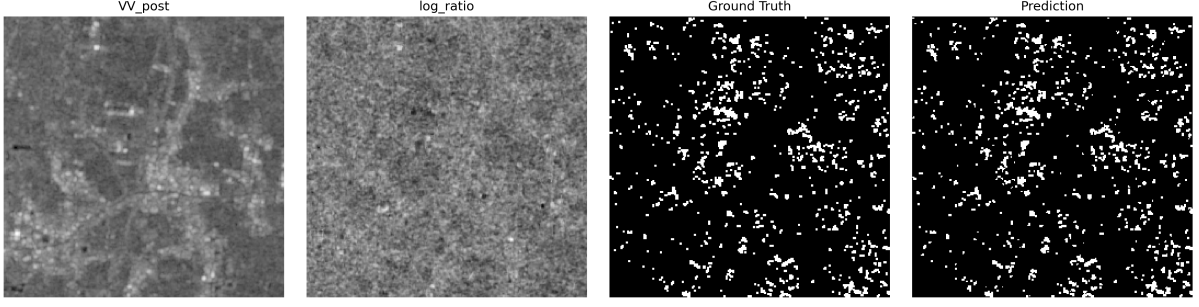


Figure 12: Qualitative comparison on a median flood-density test patch: post-flood VV backscatter, log-ratio, reference label, and network prediction (left to right). The speckled, non-contiguous character of the reference label is closely reproduced by the network prediction.

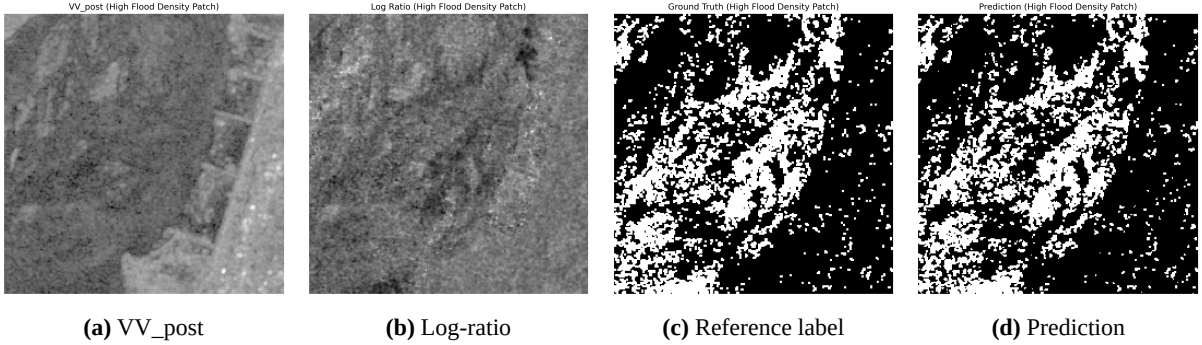


Figure 13: Qualitative comparison on the highest flood-density test patch. The network prediction closely tracks the irregular, dendritic geometry of the reference flood extent.

as one of the network’s six input channels. This is a common practice in SAR flood-mapping studies that lack access to independently surveyed ground truth (aerial photography, gauge records, or crowd-sourced reports), and it is defensible as a practical necessity, but it means the reported metrics measure the network’s ability to reproduce a specific, threshold-derived construction of flood extent, refined by morphological denoising, rather than an independently verified physical ground truth. The label-informed baseline of Section 3.11 makes this concrete: a single number, -3 dB, applied directly to the same band the labels were built from, already recovers a Dice of 0.8129 with near-perfect recall. The network’s incremental contribution beyond that baseline is real and substantial, evidenced by the precision gain from 0.6852 to 0.9591, but the absolute Dice score of 0.9593 should not be read as an independent validation of physical flood detection accuracy; it is more accurately described as a measure of how faithfully the network reproduces a specific, radiometrically motivated but ultimately heuristic labelling rule, at a spatial fidelity well beyond what that rule achieves when applied directly.

4.6 Other limitations

The study is confined to a single flood event over a single basin, and the reported metrics should not be extrapolated to other rivers, other SAR incidence geometries, or other seasons without independent validation; Sen1Floods11-scale benchmarks spanning multiple biomes and events remain the appropriate resource for claims of general applicability [8]. The train/validation/test split, while stratified by flood density, was performed at the patch level over a non-overlapping spatial grid; because immediately adjacent patches can share correlated terrain, land cover, and hydrological context, some degree of spatial autocorrelation between partitions cannot be ruled out, and a spatially blocked cross-validation scheme, withholding entire contiguous regions of the study area rather than individual patches, would offer a stronger test of generalisation within the basin. Finally, the elevation-based flat-terrain mask (below 200 m) and the coarse, twenty-day compositing windows used for the pre- and post-flood SAR composites

Table 4: Comparison of threshold-based baselines against the trained U-Net on the test partition. TB = threshold baseline; the label-informed variant is the strongest of the three and is the one plotted against the U-Net in Figure 14.

Method	Threshold	Dice	IoU	Precision	Recall	Accuracy
TB: per-patch Otsu	local, variable	0.2904	0.1699	0.1699	1.0000	0.5541
TB: global Otsu	+0.24 dB	0.2676	0.1545	0.1545	1.0000	0.5006
TB: label-informed (Youden’s J)	−3.00 dB	0.8129	0.6848	0.6852	0.9992	0.9580
U-Net (this study)	learned, spatial	0.9593	0.9218	0.9591	0.9600	0.9925

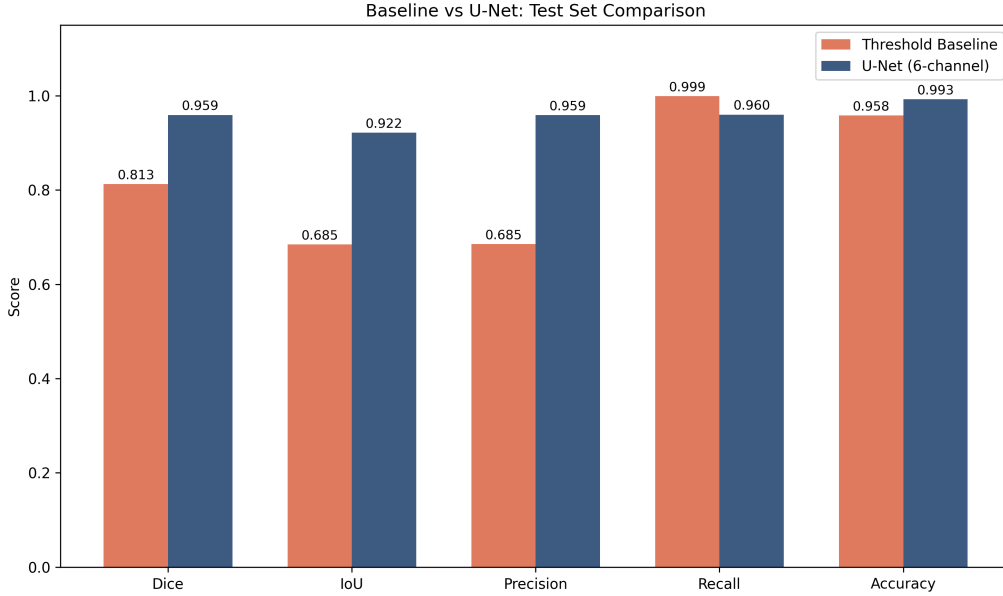


Figure 14: Comparison of the strongest (label-informed) threshold baseline against the trained U-Net across five test-set metrics. The U-Net outperforms the baseline on every metric except recall, where the baseline’s near-total flood coverage carries a substantial precision cost visible in Table 4.

both trade temporal and spatial precision for radiometric stability; a shorter compositing window closer to the flood peak, where feasible given Sentinel-1’s six-day revisit at this latitude, would sharpen the change-detection signal at some cost in speckle suppression.

4.7 Operational deployment

To make the trained checkpoint usable beyond the training notebook, a lightweight inference service was built around the exported model weights, accepting a co-registered six-band GeoTIFF matching the input stack of Table 1, padding or cropping it to the network’s 256×256 patch size, and returning the predicted flood mask alongside a summary flood-coverage percentage. The service is implemented as a Gradio application and deployed as a public inference demonstrator at <https://huggingface.co/spaces/satwikshreshth1/kosi-flood-sar-segmentation> [21], allowing a user to upload a single patch-sized, six-band SAR stack and obtain a predicted flood mask without local GPU access or direct interaction with the training code. This deployment is intended as a demonstrator of the inference path rather than an operational flood-warning system: it inherits every limitation discussed above, has been exercised only on patches drawn from the same study area and event used for training, and would require independent revalidation, ideally against surveyed ground truth, before any operational use.

5 Conclusion and Future Work

This paper presented a complete pipeline for pixel-wise flood segmentation over the Kosi river basin in Bihar, India, from raw bi-temporal Sentinel-1 SAR acquisition through Earth Engine compositing, patch-level dataset curation with explicit label-noise treatment, and the training of a six-channel U-Net under a combined cross-entropy and Dice objective. The trained network reached a test-set Dice coefficient of 0.9593 and an IoU of 0.9218 on the August–September 2024 flood pulse, substantially outperforming a label-informed single-band threshold baseline (Dice 0.8129) evaluated under identical conditions. The gap between these two figures is attributable to a combination of spatial context, multi-band radiometric information, and the network’s evident capacity to reproduce the specific denoising applied to its training labels, and the paper has argued that this last factor, together with the shared derivation of the reference labels and one input channel from the same log-ratio quantity, should temper a literal reading of the headline accuracy as an independent measure of physical flood-detection skill. Within those bounds, the results are consistent with the broader deep-learning SAR flood-mapping literature in showing a clear, reproducible advantage of a spatially aware, multi-channel U-Net over classical thresholding for this basin and event. Future work should prioritise independent validation against surveyed or optically derived ground truth where cloud conditions permit, extension to additional Kosi flood events to assess temporal generalisation, and a spatially blocked evaluation protocol to more rigorously test generalisation within the basin.

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